AURELIA

Four-voice generative MIDI sequencer for REAPER. Fourteen algorithmic composition engines, from Xenakis to the Berlin School.

User manual · version 1.3

Eloy Platas · 9 November 2026

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What Aurelia is

Aurelia is a four-voice generative MIDI sequencer for REAPER, written in JSFX. It brings together 14 algorithmic composition engines, from Xenakis to the Berlin School, in a single effect.

Each voice is a small automatic composer. It has its own engine (mode), its own MIDI output channel and its own random seed. It can play notes or, with the **CC** button, send a continuous MIDI controller to modulate filters, effects or any other synthesizer parameter.

The four voices can be independent or listen to each other. Three switches per voice decide how much: **ESC** fits them to a common scale, **RIT** to a common rhythmic grid, and **ARM** limits their harmonic distance from a guide voice. With everything off, each voice is a free process; with everything on, the four form an ensemble that respects scale, pulse and harmony.

Aurelia is designed for three uses:

- **Composing**: generate material and record the MIDI output on a track to edit it later.
- **Playing live**: transpose everything from a keyboard, switch memories and presets, or teach phrases to the Markov and intervallic engines.
- **Studying**: the 256 presets rebuild historical techniques of aleatoric, serial, stochastic, minimalist and Berlin School music, and the manual explains where each one comes from.

Everything is deterministic: the same seed and the same settings always give the same music. Changing a seed gives another variation with the same character.

History and foundations

Aurelia brings together seventy years of music generated by rules and by chance. Each mode translates a concrete idea from that history into MIDI: it does not imitate a sound, but a method of composition.

The main lines

- **Stochastic music (Xenakis, from 1954).** Iannis Xenakis used the laws of probability to compose masses of sound: clouds of notes whose attacks follow the Poisson law and whose pitches follow Gauss, Cauchy or the linear interval law. In *Analogique A* (1958-59) he used Markov chains between "screens" of sound. In his last works, *GENDY3* (1991) and *S.709* (1994), the GENDYN algorithm generates the waveform directly from a polygon whose vertices move by random walks. Aurelia brings these ideas to note level in the STOCHASTIC (ESTOCÁSTICO), GENDYN and ANALOGIQUE A modes.
- **Chance and indeterminacy (Cage, Feldman, Brown, Lutosławski, 1950-60).** John Cage made decisions with the *I Ching* in *Music of Changes* (1951). Earle Brown and Morton Feldman wrote graphic scores, and Witold Lutosławski invented "controlled aleatoricism", with repeated cells that are not synchronized. Aurelia does not use chance as noise, but as a way of choosing among possibilities fixed by the composer.
- **Serialism (Schoenberg, Webern, Boulez, Stockhausen).** The twelve-tone row and its forms (prime, retrograde, inversion, retrograde inversion), and later integral serialism, which also orders durations and dynamics. Gottfried Michael Koenig took serialism to the computer with *Project 1* (1964) and *Project 2* (1966): the composer chooses a degree of regularity between the strict row and chance. This is the SERIAL mode.
- **Combinatorics and computer-assisted composition (OpenMusic, IRCAM).** Permutations, rotations and combinations of a row, and the search for solutions that satisfy constraints (OMClouds, Situation). These are the SERIAL and REGLAS (rules) modes.
- **Fractals and self-similarity (Nørgård, Voss, Lindenmayer).** Per Nørgård discovered his "infinity series" around 1959, a melody that contains slow copies of itself. Richard Voss and John Clarke showed in 1975 that music has a spectrum close to 1/f. Lindenmayer systems, grammars that rewrite themselves, describe plants and also melodies. This is the FRACTAL mode.
- **Cellular automata (Wolfram, Otomata).** Simple local rules that produce order, chaos or complexity, such as Stephen Wolfram's 256 elementary rules or the musical toy Otomata (2011). This is the AUTÓMATA (automata) mode.
- **Minimalism and process (Riley, Reich, Glass, Eno, Pärt).** Repetition, phasing, gradual process, loops of incommensurable lengths. It has no mode of its own, but many presets rebuild it with TURING, BLOOM and EUCLÍDEO (Euclidean).
- **Berlin School (Tangerine Dream, Klaus Schulze, 1970s).** Analog 8- or 16-step sequencers in a loop, slowly moving filters and long pads. The TURING and ESPACIAL (spatial) modes, and the CC output, come from this tradition.
- **Modern modular sequencers.** Music Thing's Turing Machine, Qu-Bit's Bloom, Make Noise's René, monome's Kria, Mutable Instruments' Grids, the Torso T-1, Hemisphere, Eventide Misha and the Samchillian keyboard. Aurelia takes their control ideas, not their code.
- **Euclidean rhythm (Toussaint, 2005).** Godfried Toussaint showed that Euclid's algorithm, spreading k hits over n steps as evenly as possible, generates many of the world's traditional rhythms. This is

the EUCLÍDEO mode.

Design principles

1. **Each voice, a composer.** Four voices, each with its engine, its seed and its channel.
2. **The relationship between voices is a choice.** ESC, RIT and ARM decide how much they listen to each other, from total independence to a harmonic ensemble.
3. **Determinism.** All randomness comes from seeded pseudo-random generators: the same configuration always gives the same music, and recordings are reproducible.
4. **Explain, don't hide.** Every control has its help in the interface, every mode a view that shows its internal state, and every preset says where it comes from.

Versions

Version	What's new
1.3	CC output per voice with selectable number, 256 presets and this manual
1.2	Seven new families: SERIAL, FRACTAL, AUTÓMATA, ESPACIAL, EUCLÍDEO, REGLAS, INTERVÁLICO
1.1	Transposition from a MIDI keyboard and Markov engines: ANALOGIQUE A, SUBSEQ, MARKOV
1.0	Four voices with ESTOCÁSTICO, GENDYN, TURING and BLOOM, and an interface in the style of GenderBender P23

A note on names

The interface of Aurelia is in Spanish. This manual keeps the names exactly as they appear on screen (in capitals) and translates them the first time they appear. The most frequent ones: ESC = scale, RIT = rhythm, ARM = harmony, GUÍA = guide, MODO = mode, TIPO = type, ESTOCÁSTICO = stochastic, ESPACIAL = spatial, EUCLÍDEO = Euclidean, REGLAS = rules, INTERVÁLICO = intervallic, AUTÓMATA = automaton, DIV = step division, BASE = lowest note, RANGO = range, PROB = probability.

Installation and first steps

Aurelia is a single file, `aurelia.jsfx`. You install it by copying it into REAPER's effects folder and use it like any other JS effect on a MIDI track.

Install

1. Copy `aurelia.jsfx` (and, if you like, `README.md`) into `~/.config/REAPER/Effects/Aurelia/` on Linux, `%APPDATA%\REAPER\Effects\Aurelia\` on Windows or `~/Library/Application Support/REAPER/Effects/Aurelia/` on macOS.
2. In REAPER, open the effects browser and press F5 so it finds it. It appears as **JS: Aurelia**.
3. REAPER 6.74 or later is required, because Aurelia uses more than 64 parameters.

First sound in three minutes

1. Create a track, insert **JS: Aurelia** and leave the track without audio input.
2. Create four tracks with synthesizers (or one multitimbral synthesizer) receiving MIDI from Aurelia's track on channels 1, 2, 3 and 4. In REAPER this is done with a MIDI send from Aurelia's track, filtering the channel on each receiving track. If you are going to use drum presets, add another track with a General MIDI drum kit on channel 10.
3. Press play in REAPER. With the clock on **HOST**, Aurelia follows the transport; with **LIBRE** (free), it plays even when the transport is stopped.
4. Open the **PRESET** menu (click on the name) and try, for example, *129 · Phaedra* or *108 · Music for Airports*.
5. Hover over any control: the help box on the right explains what it does.

Recording what it generates

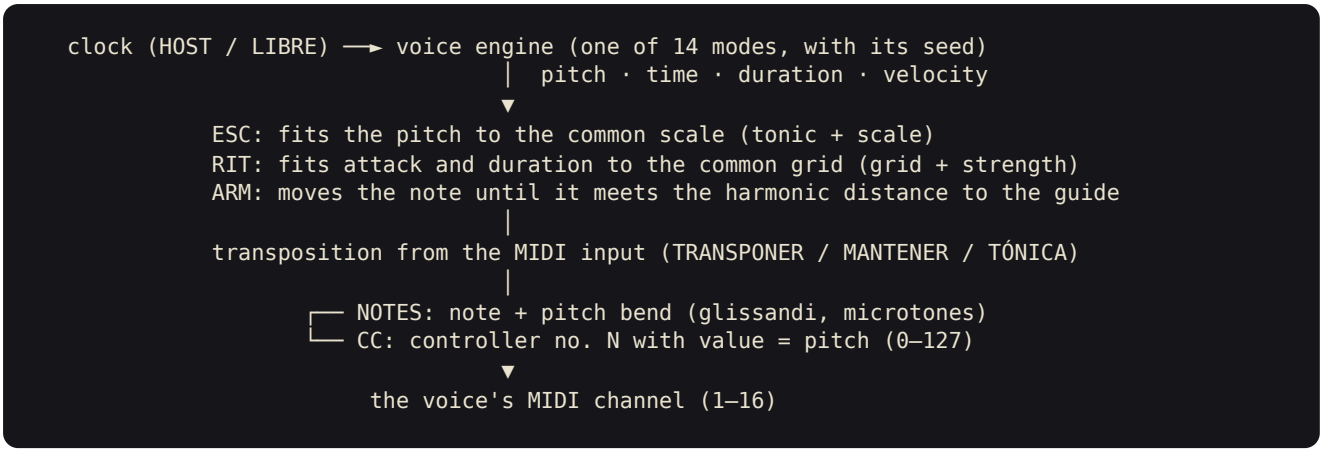
Aurelia produces MIDI, not audio. To keep a take, create a track that receives the MIDI output of Aurelia's track, arm it in *Record: MIDI output* mode and record. You get an ordinary MIDI item that you can edit.

If there is no sound

- **The transport is stopped and the clock is on HOST.** Press play or switch the clock to LIBRE.
- **The voice is off** (ON button) or in **CC** mode: in CC there are no notes, only controllers.
- **The channels do not match.** Each voice shows its channel under MIDI OUT. The presets use channels 1 to 4, channel 10 for drums and the modulated voice's channel for CCs.
- **A very sparse cloud** (for example *Silencio estructurado*, "structured silence") may take a while to sound. That is intentional.

Architecture

Each voice generates events with its engine; they then go through three common filters (scale, grid and harmony) and leave as notes or as CC on the voice's channel.



Clock

- **HOST**: Aurelia reads REAPER's position and tempo. On play, if **RESET AL PLAY** (reset on play) is active, all voices go back to the beginning and the music repeats exactly.
- **LIBRE** (free): an internal clock with its own **TEMPO**; it plays with the transport stopped, useful live.
- **HOST o LIBRE** (host or free): follows REAPER while it plays and uses the internal clock when it is stopped.

Events are computed in quarter-note beats and scheduled with sample accuracy inside each audio block, so the rhythm does not depend on the buffer size.

The three quantizations

Switch	What it does	Global settings
ESC	Moves each pitch to the nearest note of the common scale. Off, the voice is chromatic and can be microtonal.	TÓNICA (tonic), ESCALA (scale, 27 scales)
RIT	Moves attacks and durations to the common grid. With FUERZA (strength) below 100 % it only pulls them closer.	REJILLA (grid), FUERZA
ARM	Limits the Tenney harmonic distance from the guide voice (or from all voices): the note moves to the nearest one that meets the limit.	DIST ARM, GUÍA (guide), contra

The **Tenney harmonic distance** of a just interval p/q is $\log_2(p \cdot q)$. It is 0 for the unison, 2.58 for the fifth, 3.58 for the fourth, 3.91 for the major sixth, 4.32 for the major third, 4.91 for the minor third, and rises to 10.49 for the tritone. Each octave adds 1. With DIST ARM = 2.6 only unisons, fifths and octaves are allowed against the guide; with 4.4, also major thirds, sixths and fourths.

Notes or CC

With the **CC** button off, the voice sends notes. Glissandi (GENDYN, GLISS, SLIDE) and microtones use pitch bend, so the global **BEND** must match your synthesizer's pitch-bend range.

With **CC** on, the voice sends the chosen controller (0–127). Its value is the pitch that would have sounded, already adjusted by ESC and ARM. There are no notes and no pitch bend; the value is sent on each attack and held until the next one, like a *sample & hold*. For finer movement, use a fast grid or division.

To modulate another voice's synthesizer, put the CC voice on the same channel. The CC range depends on the engine's register (CENTRO and ANCHO, BASE and RANGO...). For example, GENDYN with CENTRO C5 (72) and ANCHO 48 covers almost the whole range, from 24 to 120.

MIDI input

Notes arriving at Aurelia's track have three uses:

1. **Transposition**: TRANSPONER (transpose), MANTENER (hold) or TÓNICA (tonic), relative to the **REF** note.
2. **Markov corpus**: the MARKOV mode with FUENTE (source) = ENTRADA (input) learns from what you play.
3. **Intervals**: the INTERVÁLICO mode with FUENTE = ENTRADA takes each key as an interval relative to REF.

CANAL filters the input channel and **THRU** passes the input to the output.

Interface, memories and presets

The window has four voice columns, a right-hand panel with the global settings, a memory bar at the bottom and the preset selector at the top.

Voice column (top to bottom)

1. **VOZ n, LED, ON and TEST.** The LED flashes with each note; TEST sends a test note on the voice's channel.
2. **MODO.** One of the 14 engines. Arrows, wheel or click for the menu. Changing mode loads its default values.
3. **MIDI OUT.** The channel (1–16), the **CC** button and the controller number. The selector shows "notas" (notes) when CC is off; its menu groups the 128 controllers in blocks of 16 and marks the usual ones.
4. **ESC, RIT and ARM.** The three quantizations. On the guide voice, ARM shows as **GUÍA**.
5. **Mode view.** A drawing of the engine's internal state: GENDYN's polygon, the Turing Machine register, the Otomata grid, the Euclidean rings...
6. **The 16 mode knobs**, in a 4×4 grid.
7. **Output: VEL, DIN, GATE and SEMILLA** (velocity, dynamics, gate, seed).

Using the controls

Action	Result
Drag vertically	Changes the value
Shift + drag	Fine adjustment
Wheel	Step by step
Right-click or Ctrl + click	Back to the default value
Click in the centre of a selector	Opens the menu
Hover	Shows the help in the right-hand panel

Right-hand panel

- **PARTITURA** (score): the last 8 beats of the four voices, each in its colour.
- **ESCALA**: tonic, scale and a keyboard that marks the notes of the scale and the ones sounding.
- **RITMO**: grid and quantization strength.
- **ARMONÍA**: maximum harmonic distance, guide voice and what is measured against (guide or all).
- **ENTRADA MIDI** (MIDI input): transposition mode, reference note, channel and THRU.
- **RELOJ Y SALIDA** (clock and output): clock, free tempo, pitch-bend range, global seed and reset on play.
- **Help**: the text of the control under the mouse.

Memories

Ten memories store the complete state of Aurelia (all parameters of the four voices and the globals, except the MIDI input) and are saved with the project. Click or keys 1–0 to recall; hold for 0.6 s to store. A memory lit in gold is the last one recalled.

Presets

The **PRESET** selector loads one of the 256 factory configurations, arranged in 8 categories of 32.

- Arrows or wheel to step through them; click for the menu by category.
- Hover over it to read in the help which work or technique inspired it.
- Loading a preset sets modes, parameters, scale, tonic, grid, harmony, seeds and channels, and restarts the four voices. Your clock, pitch-bend and MIDI input settings do not change.
- Channels follow a fixed convention: voices 1–4 go out on channels 1–4, General MIDI drums on 10, and CC voices on the channel of the voice they modulate.
- The **Preset cargado** (loaded preset) parameter can be automated in REAPER to switch presets during a piece. When a project is opened it is not reloaded: the saved settings are kept.
- To keep a variation of a preset, use a memory or REAPER's effect presets (the + button in the effect window).

Global and common voice parameters

Global parameters affect the four voices; common parameters exist in every voice whatever its mode. The parameter number is the one REAPER shows for automation.

Globals (right-hand panel)

No.	Control	Values	What it does
1	RELOJ (clock)	HOST, LIBRE, HOST o LIBRE	Where time comes from (see Architecture)
2	TEMPO	20–300 BPM	Tempo of the free clock
3	TÓNICA (tonic)	C–B	Tonic of the common scale
4	ESCALA (scale)	27 scales	Scale used by voices with ESC
5	REJILLA (grid)	1/64 to 4/1 (with triplets and dotted)	Common grid for voices with RIT
6	FUERZA (strength)	0–100 %	100 % snaps to the grid; less only pulls closer
7	DIST ARM	0–12	Maximum Tenney distance; 2.6 = up to the fifth, 4.4 = thirds and sixths, 12 = no limit
8	GUÍA (guide)	voice 1–4	Harmonic reference for voices with ARM
9	contra (against)	GUÍA, TODAS	Measure against the guide only, or against the last note of every voice
10	BEND	1–48 semitones	Pitch-bend range of your synthesizers
11	RESET AL PLAY	Yes / No	Go back to the start of the sequence on play
12	SEMILLA (seed)	0–999	Changes the random material of the four voices
13	TRANSP	OFF, TRANSPONER, MANTENER, TÓNICA	Transposition from the MIDI input
14	REF	C1–C7	Input note that does not transpose (C4 by default)
15	THRU	Yes / No	Pass the input to the output
16	CANAL (input channel)	OMNI, 1–16	Channel Aurelia listens to
17	Preset cargado	0–256	Current preset; changing it loads the preset

Available scales: Major, Natural minor, Harmonic minor, Melodic minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian, Major pentatonic, Minor pentatonic, Blues, Whole tone, Octatonic H-W, Octatonic W-H, Messiaen 3, Augmented, Hungarian minor, Phrygian dominant, Acoustic, Prometheus, Enigmatic, Hirajoshi, In sen, Pelog (approx.), Double harmonic and Chromatic.

Common to each voice

Voice n occupies the parameters from $21 + 30\cdot(n - 1)$. The offsets are:

Offset	Control	Values	What it does
+0	ON	Yes / No	Switches the voice on or off
+1	CANAL (channel)	1–16	MIDI output channel
+2	MODO (mode)	0–13	The voice's engine
+3	ESC	Yes / No	Fit to the common scale
+4	RIT	Yes / No	Fit to the common grid
+5	ARM	Yes / No	Limit the harmonic distance to the guide
+6	VEL	1–127	Base velocity. In ANALOGIQUE A and MARKOV it acts as a percentage of the dynamics the engine generates
+7	DIN	0–100 %	Random velocity variation
+8	GATE	5–200 %	Note length relative to its step; 100 % or more is legato (needed for continuous glissandi)
+9	SEMILLA (seed)	0–999	The voice's seed: the same seed always gives the same material
+10 to +25	P1–P16	depends on the mode	The mode's 16 knobs (see the mode chapters)
+26	SALIDA (output)	Notes / CC	Notes or continuous controller
+27	CC NO.	0–127	Controller sent in CC mode

Units you will see on the knobs

- **t** = quarter-note beats. *0.5 t* is an eighth note.
- **st** = semitones; **ct** = cents.
- **oct** = octaves above BASE or RAÍZ (root).
- **Divisions**: 1/16 is a sixteenth note, 1/8T an eighth-note triplet, 1/8. a dotted eighth, 2/1 two whole notes.
- Notes follow REAPER's convention: C4 = 60 (middle C), C3 = 48.

Stochastic and Markov modes

The first seven modes come from Xenakis's stochastic music, from modular sequencers and from Markov chains. In each table, the *Range* column gives the limits of the knob.

0 · ESTOCÁSTICO (stochastic)

History. In *Pithoprakta* (1956) and *Achorripsis* (1957), Xenakis composed clouds of sound with probability laws: attack times follow a Poisson process (exponential intervals), pitches follow a chosen law, and the overall form is a succession of sections with different densities. The linear interval law, f(x) = 2(1 − x), makes small intervals more likely.

How it works. Each event is scheduled after an exponential interval whose mean is 1/DENS. The pitch comes from the chosen LEY (law) around CENTRO (or from the previous note if PASEO is high). Density and centre drift slowly. Sections (SECC) decide by coin toss whether the voice plays or rests.

Knob	Range	What it does
DENS	0.05–8 ev./t	Mean number of events per beat (Poisson)
DERIVA (drift)	0–100 %	Density wanders: the cloud thickens and thins
LEY (law)	8 laws	UNIF, LINEAL (Xenakis), GAUSS, CAUCHY, LOGIST, COSH, ARCSEN, EXPON
FORMA (shape)	0–100 %	Concentration or tail of the law
CENTRO (centre)	C1–C8	Central pitch of the cloud
ANCHO (width)	1–48 st	Width of the register on each side
DERIVA C	0–100 %	The centre moves through the register
PASEO (walk)	0–100 %	0 = independent pitches; 100 = random walk from the previous note
MICRO	0–100 ct	Microtonal deviation with pitch bend (with ESC off)
DUR	0.05–4 t	Mean duration (exponential)
NUBE (cloud)	1–6	Simultaneous notes per event
GLISS	0–100 %	Probability of arriving with a glissando
SECC (sections)	OFF, 1–32 bars	Mean length of the sections
ACTIVO (active)	0–100 %	Probability that a section plays

1 · GENDYN

History. Dynamic stochastic synthesis was Xenakis's last project, with *GENDY3* (1991) and *S.709* (1994). A waveform is a polygon of N vertices; on each cycle, the amplitude and duration of each vertex move by a random walk with reflecting barriers. Peter Hoffmann rebuilt the program and showed that the position of the barriers is what changes the result most.

How it works. Aurelia applies GENDYN at note level: each vertex is a note and its segment the time until the next one. With ORDEN (order) 2, the steps move a velocity and the velocity moves the point, with more inertia (as in GENDY3). GLISS joins the vertices with pitch bend, like the polygonal wave.

Knob	Range	What it does
PUNTOS (points)	3–32	Polygon vertices (notes per cycle)
ORDEN (order)	1–2	Simple or second-order walk
LEY A / LEY D	6 laws	Laws of the amplitude and duration steps: LINEAL, CAUCHY, LOGIST, COSH, ARCSEN, EXPON
PASO A / PASO D	0–100 %	Step size
FORMA	1–100 %	Parameter *a* of the laws
BARRERA (barrier)	0–100 %	Opening of the amplitude barriers; 0 freezes the pitch
CENTRO	C1–C8	Pitch of amplitude 0
ANCHO	1–48 st	Semitones of amplitude ±1
T MIN / T MAX	0.03–4 t	Barriers of a segment's duration
PERIODO (period)	LIBRE, 1–32 t	Free cycle (Gendy1) or fixed-length cycle (GENDY3)
GLISS	OFF / ON	Interpolation with pitch bend
SECC / ACTIVO	as in ESTOCÁSTICO	Sections that play or rest

2 · TURING

History. Music Thing Modular's Turing Machine (Tom Whitwell, 2012) is a 16-bit shift register. On each step the bit that leaves comes back in, inverted or not according to the probability set by the big knob. At the top, the loop is locked; in the middle everything is random; at the bottom, the loop inverts on each pass and lasts twice as long.

How it works. The first 8 bits form a number from 0 to 255 (the DAC), scaled to RANGO semitones above BASE and fitted to the scale if ESC is on.

Knob	Range	What it does
BLOQUEO (lock)	0–100 %	100 = locked loop; 50 = random; 0 = inverted loop (x2)
LONG (length)	2–16	Loop length
RANGO (range)	1–48 st	Semitones covered by the DAC
BASE	C1–C7	Lowest note
DIV	1/64–4/1	Length of each step
PULSOS (pulses)	TODOS, BIT 1–8	A note on every step, or only when that bit is 1 (Pulses expander)
SLIDE	0–100 %	Probability of a glide from the previous note

3 · BLOOM

History. Bloom, by Qu-Bit Electronix (2020), is a fractal sequencer: a trunk of up to 32 steps generates branches through successive transformations, and the user chooses the path through the tree.

How it works. After the trunk, RAMAS (branches) variants play; each derives from the previous one by transposition, inversion, retrograde or mutation, according to the CAMINO (path, one of 128). **NUEVO** (new) generates another trunk and **DESH.** (undo) recalls the previous one.

Knob	Range	What it does
LONG	1–32	Trunk steps
RAMAS (branches)	0–7	Branches after the trunk
CAMINO (path)	0–127	Route through the tree
MUTAR (mutate)	0–100 %	Changes in the trunk at the end of each pass
RAÍZ (root)	C1–C7	Root note
RANGO	1–4 oct	Register of the trunk
ORDEN (order)	ADELANTE, ATRÁS, PÉNDULO, AZAR	Forward, backward, pendulum, random
DIV	1/64–4/1	Length of each step
SILENC	0–100 %	Silent steps
RATCH	0–100 %	Ratchet repeats (2 to 4)
SLIDE	0–100 %	Steps that arrive with a glide

4 · ANALOGIQUE A

History. In *Analogique A* for 9 strings (1958-59), Xenakis organized sound into "screens": snapshots defined by pitch region (f), dynamics (g) and density (d). The succession of screens follows a Markov chain with an 8×8 transition matrix, at equilibrium or perturbed. Aurelia follows the reconstructions by Agostino Di Scipio and Gonçalo Carvalho.

How it works. Every ΔT a screen (A–H) is chosen and filled with grains: 1, 3 or 9 events per cell, in six pitch regions from E1 to A6, at quintuplet, quadruplet or triplet positions. VEL acts as a percentage of the screen's dynamics.

Knob	Range	What it does
ΔT	0.25–8 t	Length of each screen
MODELO (model)	GTPM, ACOPLADO	The work's 8×8 matrix, or three coupled two-state chains with alpha/beta matrices
PERTURB	0–100 %	Forces the DESTINO screen (with PROTOC. off)
DESTINO (target)	A–H	Screen the perturbation tends towards
PROTOC. (protocol)	OFF / ON	Sections of 30 screens, at equilibrium or perturbed, as in the work
DENS	10–300 %	Scale of the densities

Knob	Range	What it does
REGISTRO (register)	−3 to +3 oct	Transposition of the regions
SUBDIV	5+4+3, LIBRE	Grid of quintuplets, quadruplets and triplets, or free time
DUR	5–100 %	Length of the long notes relative to the screen
LEGNO	0–100 %	Proportion of very short notes (col legno)

5 · SUBSEQ

History. Inspired by the Python library *Subsequence*, which generates music from a weighted chord graph, Euclidean rhythms and voice leading. It adds Eugene Narmour's melodic inertia: after a leap, the melody tends to change direction.

How it works. A progression walks the GRAFO (graph) with attraction to the tonic. Each voice plays a PAPEL (role) over it with a Euclidean rhythm of GOLPES (hits) in 16 steps, rebuilt each cycle and thinned out as the tension of the 4-chord section grows.

Knob	Range	What it does
PAPEL (role)	ACORDES, BAJO, MELODÍA, ARPEGIO	Chords, bass, melody or arpeggio
GRAFO (graph)	FUNC., MODAL, CROMÁT., PEDAL	Kind of progression (tonal, neighbouring degrees, P/L/R, always back to the tonic)
GRAVED. (gravity)	0–100 %	Attraction to the I chord
CAMBIO (change)	1–16 t	Length of each chord
FORMA (form)	LIBRE, FIJA, AABA	Live progression, frozen, or frozen A and live B
BASE	C1–C7	Lowest note of the voice
DIV	1/64–4/1	Length of each rhythm step
GOLPES (hits)	1–16	Hits of the Euclidean rhythm per cycle
TENSIÓN	0–100 %	The rhythm thins out as the section advances
INERCIA	0–100 %	Narmour inertia (MELODÍA)
VARIA	0–100 %	Probability of rotating the rhythm
NOTAS	3–4	Triads or seventh chords

6 · MARKOV

History. Markov chains have been used in composition since Hiller and Isaacson's *Illiac Suite* (1957) and *Analogique A*. This mode follows the Max for Live devices that learn from what you play: each new note is chosen by looking at what followed the last N notes in the learned corpus.

How it works. The corpus comes from the MIDI input, from the other voices or from a random melody. If there are no matches for the requested order, the chain drops to a lower order. Rhythm has its own chain.

Knob	Range	What it does
FUENTE (source)	ENTRADA, VOCES, AZAR	Learns from the input, the voices or a random corpus
ORDEN (order)	1–4	Previous notes it looks at
ORDEN R	0–4	Order of the rhythm chain (0 = fixed rhythm with DIV)
MODO	ABS., INTERV.	Learns notes or intervals
TEMPER	0–100 %	Probability of jumping to any note of the corpus
VENTANA (window)	4–512	Recent notes used
CONGELA (freeze)	OFF / ON	Stops learning
DIV	1/64–4/1	Step when ORDEN R = 0
TRANSP	±24 st	Transposes the output
SILENC.	0–100 %	Notes that are skipped
REINIC. (restart)	0–100 %	Break the chain and jump to another point in the corpus

Serial, fractal, automata, spatial, Euclidean, rules and intervallic modes

In these seven modes the first knob, **TIPO** (type), chooses the specific system. Some knobs only affect certain types; the table says so. BASE is always the lowest note and RANGO the octaves above it; pitches that go out of range bounce back inside.

7 · SERIAL (serial and combinatorics)

History. Schoenberg established the twelve-tone technique in the 1920s, with its forms P (prime), R (retrograde), I (inversion) and RI. Messiaen, Boulez and Stockhausen extended the row to durations and dynamics. Koenig took the idea to the computer:

- **Project 1** (1964): each parameter has a level from 1 to 7, from strictly regular (row in order, repeated groups) to irregular (chance without immediate repetition).
- **Project 2** (1966): the composer chooses a selection method for each parameter: ALEA, SERIE, SEQUENCE, GROUP, TENDENCY (a mask that opens and closes) or RATIO.

Eventide Misha reads sequences with serial forms, and OpenMusic offers rotations, permutations and combinations.

How each type works.

- **PR1** and **PR2**: three independent selectors for pitch (from the row), rhythm (1–6 units) and dynamics (6 levels).
- **MISHA**: the row in the chosen FORMA, plus SHIFT degrees on each pass.
- **ROTAR** (rotate): each pass starts one step further on.
- **PERMUTAR** (permute): each pass is the next lexicographic permutation.
- **COMBINAR** (combine): runs through the combinations of half the row.

Knob	Range	What it does
TIPO	PR1, PR2, MISHA, ROTAR, PERMUTAR, COMBINAR	System
LONG	3–12	Row length; 12 with ESC off = twelve-tone
FORMA	P, R, I, RI, CICLO, AZAR	Reading form (MISHA); CICLO cycles, AZAR random
SHIFT	±12 degrees	Progressive transposition per pass
RI ALT / RI RIT / RI DIN	1–7	PR1: regularity level of pitch, rhythm and dynamics
SEL ALT / SEL RIT / SEL DIN	6 methods	PR2: selection method of each parameter
BASE / RANGO	C1–C7 / 1–4 oct	Register
UNIDAD (unit)	1/64–4/1	Rhythmic unit (in PR1/PR2 durations are 1–6 units)

Knob	Range	What it does
TEND	1–32 bars	Cycle of the tendency mask
OCTAVA	0–100 %	Octave displacements (free register of the row)
GRUPO	1–8	Maximum size of the repetition groups

8 · FRACTAL

History. Per Nørgård found his *infinity series* around 1959: it starts at 0 and follows $a_{2n} = -a_n$ and $a_{2n+1} = a_n + 1$ (0, 1, −1, 2, 1, 0, −2, 3…). Reading one term in 2, 4 or 8, the whole series appears again, transposed or inverted. He used it in *Voyage into the Golden Screen* (1968) and in his Symphonies 2 and 3. Richard Voss proposed the dice that generate 1/f (pink) noise, and Aristid Lindenmayer the grammars that rewrite symbols in parallel (L-systems), applied to music by Prusinkiewicz in 1986.

How each type works.

- **NØRGÅRD**: term n of the series, multiplied by INTERV.
- **1/F**: the sum of DADOS (dice) random values; die k changes every 2^k steps.
- **L-SYSTEM**: rewrites the GRAMÁTICA (grammar) ITER times and reads the symbols as steps of PASO degrees. In CANTOR, B is silence; in PLANTA (plant), brackets store and recall the position.

Knob	Range	What it does
TIPO	NØRGÅRD, 1/F, L-SYSTEM	System
INTERV	±7 degrees	Interval of the Nørgård series
INICIO (start)	0–64	Entry point in the series
DADOS (dice)	2–12	Voss dice: more dice, more memory and more register
BLANCO (white)	0–100 %	White-noise mix
GRAMÁT	FIBONACCI, THUE-MORSE, CANTOR, PLANTA	L-system grammar
ITER	1–8	Rewriting generations
PASO (step)	±4 degrees	Movement of each symbol
BASE / RANGO / DIV		Register and rhythm step
RIT FRAC	0–100 %	Lengthens notes x2 or x3 according to another reading of the Nørgård series
PROB	0–100 %	Probability that each step plays

9 · AUTÓMATA (automaton)

History. An elementary cellular automaton is a row of live or dead cells; each cell looks at its two neighbours and decides its state according to an 8-bit rule. Stephen Wolfram classified the 256 rules in the 1980s (rule 30 is chaotic, 90 draws the Sierpinski triangle, 110 is universal). *Otomata* (Batuhan Bozkurt, 2011) is a musical toy of cells bouncing in a grid.

How each type works.

- **OTOMATA**: CELDAS (cells) move through a 9×9 grid. When they hit a wall they sound (their row or column gives the degree) and turn back; if two meet, they rotate clockwise.
- **ECA-256**: four rows of 8 cells with rule REGLA + row × VARIA; each birth is a note.
- **WOLFRAM**: one row of 16 cells read according to LECTURA (reading).

An automaton that dies out is reseeded with DENS INI.

Knob	Range	What it does
TIPO	OTOMATA, ECA-256, WOLFRAM	System
REGLA (rule)	0–255	Wolfram rule
VARIA	0–64	ECA-256: rule difference between rows
CELDAS (cells)	1–16	OTOMATA: number of cells
DENS INI	0–100 %	Density when reseeding
MUTAR (mutate)	0–100 %	Flips a random cell on each step
POLI	1–4	Simultaneous notes per step
LECTURA	BINARIA, CUENTA, CENTRO	WOLFRAM: the centre as a binary number, the number of live cells, or only if the centre cell is alive
BASE / RANGO / DIV		Register and step

10 · ESPACIAL (spatial, geometric)

History. Three ideas of sequencing in space:

- **René** (Make Noise, 2013): a 4×4 Cartesian sequencer with X and Y clocks and a third dimension, Z.
- **Kria** (monome): each parameter (trigger, note, octave, duration) has its own loop of a different length.
- **Orca** (Hundred Rabbits, 2018): a language of operators on a grid; Aurelia takes its clock, increment and random operators.

How each type works.

- **RENÉ**: the cursor moves through the grid according to RECORRIDO (path); each cell has a pitch and a trigger.
- **KRIA**: four loops of lengths L TRIG, L NOTA, L OCT and L DUR that advance together.
- **ORCA**: LÍNEAS (lines) operators, each with its own period, step, modulus and offset.

Knob	Range	What it does
TIPO	RENÉ, KRIA, ORCA	System
DIV X	1/64–4/1	Main clock
DIV Y	1/64–4/1	RENÉ: Y clock (with RELOJ XY)

Knob	Range	What it does
RECORRIDO (path)	FILAS, ZIGZAG, RELOJ XY, DIAGONAL, AZAR	RENÉ: rows, zigzag, independent XY clocks, diagonal, random
PÁGINAS (pages)	1–4	RENÉ: Z pages of 16 cells
GATES	0–100 %	Cells with an active trigger
MUTAR	0–100 %	Changes cells (in ORCA, random notes)
BASE / RANGO		Register
L TRIG / L NOTA / L OCT / L DUR	1–16	KRIA: length of each loop
LÍNEAS (lines)	1–8	ORCA: number of lines
PROB	0–100 %	Probability that a trigger plays

11 · EUCLÍDEO (Euclidean, distribution)

History. Godfried Toussaint (2005) showed that E(k, n), k hits spread over n steps as evenly as possible (the Bjorklund algorithm), produces traditional rhythms from all over the world: E(3,8) is the tresillo, E(5,16) the bossa nova and E(7,12) the bembé bell. Modern modules combine it with other ideas: the Torso T-1 with melody and repeats, Grids (Mutable Instruments) with an interpolated map of drum patterns, and Hemisphere with logic between rings.

How each type works.

- **TOUSSAINT**: ring A gives the rhythm; ring B decides when the melody of NOTAS pitches advances.
- **T-1**: ring A, with melody, REPEAT repeats, ARMONÍA notes per hit and CAOS (chaos).
- **GRIDS**: three lanes (kick, snare, hi-hat) read from a 2D map at the X/Y position; they play if they exceed the DENS threshold. With NOTAS = 1 it sends GM notes 36, 38 and 42; switch ESC and ARM off on that voice.
- **ANULAR** (ring): rings A and B are combined with LÓGICA (logic).

Knob	Range	What it does
TIPO	TOUSSAINT, T-1, GRIDS, ANULAR	System
PASOS / PULSOS / ROT	1–32 / 0–32 / 0–31	Ring A: steps, pulses, rotation
PASOS2 / PULSOS2	1–32 / 0–32	Ring B
LÓGICA	OR, AND, XOR, A→NOTA	ANULAR: how the rings combine
X / Y	0–100 %	GRIDS: position on the map
DENS	0–100 %	GRIDS: density
CAOS (chaos)	0–100 %	GRIDS and T-1: random perturbation
NOTAS	1–8	Pitches of the melody
REPEAT	1–4	T-1: repeats per hit
BASE / DIV		Register and step

Knob	Range	What it does
ARMONÍA	1–4	T-1: notes stacked in thirds

12 · REGLAS (rules, constraints)

History. Constraint-based composition describes what the music must satisfy and lets a program search for solutions. In OpenMusic (IRCAM), notable examples are OMClouds, by Charlotte Truchet, with adaptive local search, and Situation, by Camilo Rueda and Antoine Bonnet. The bach library for Max follows that line. *Music Mouse* (Laurie Spiegel, 1986) is an "intelligent instrument": the mouse moves voices that obey rules of harmony. The *Illiac Suite* (1957) already generated counterpoint by applying rules.

How each type works.

- **OMCLOUDS**: a phrase of LONG notes. On each pass, ITER times, the note with the most conflicts is found and replaced by the one with the fewest. The rules are INTERV, SALTO (leap), NO REP and a triad on the strong beat; TEMP introduces perturbations.
- **SITUATION**: chords of VOCES notes with vertical interval INTERV and the smallest possible horizontal movement.
- **CONTRAP.**: first-species counterpoint against the last note of the guide voice. Consonances only, no parallel fifths or octaves, prefers contrary motion and penalizes large leaps.
- **MUS.MOUSE**: a wandering cursor (MOVIM) moves two pairs of voices in parallel, contrary or oblique motion, as chords or arpeggios.

Knob	Range	What it does
TIPO	OMCLOUDS, SITUATION, CONTRAP., MUS.MOUSE	System
LONG	4–16	OMCLOUDS: phrase length
ITER	1–32	OMCLOUDS: corrections per pass
INTERV	2nd/3rd, 3rd, 4th, LIBRE	Allowed intervals (in SITUATION, the vertical one)
NO REP	0–4	Previous notes that may not be repeated
SALTO (leap)	1–12 degrees	Maximum leap
TEMP	0–100 %	Temperature (perturbation)
BASE / RANGO / DIV		Register and step
VOCES (voices)	2–4	SITUATION and MUS.MOUSE
MOVIM	0–100 %	MUS.MOUSE: cursor speed
MOV	PARALELO, CONTRAR., OBLICUO	MUS.MOUSE: relationship between the pairs
POSICIÓN	ENCIMA, DEBAJO, LIBRE	CONTRAP.: above, below the guide, or free
PATRÓN	ACORDE, ARP SUBE, ARP BAJA, ALTERNA	MUS.MOUSE: chord, arpeggio up, down, alternating

13 · INTERVÁLICO (intervallic, relative)

History. In relative music, intervals are fixed rather than notes. The Eventide Misha (2019) is an instrument-sequencer that works with degrees and movements within a scale. The Samchillian Tip Tip Tip Cheeepeeeee, by Leon Gruenbaum (1990s), is a keyboard where each key means "go up or down so many degrees" from the previous note.

How each type works.

- **MISHA**: a sequence of LONG intervals that accumulate, read with FORMA (R reverses order and sign; I reverses the sign) and shifted by SHIFT at the start of each pass.
- **SAMCHILL**: each note comes from the previous one by an interval of up to MAX INT degrees; PESO (weight) favours small ones.

With **FUENTE = ENTRADA** (source = input), the keys you play are intervals relative to REF: D4 goes up a tone, B3 down a semitone. MISHA records them in its sequence and SAMCHILL plays them immediately. Set the global transposition to OFF so they do not add up.

Knob	Range	What it does
TIPO	MISHA, SAMCHILL	System
FUENTE (source)	AUTO, ENTRADA	Generated or played intervals
LONG	2–32	MISHA: sequence length
FORMA	P, R, I, RI, CICLO, AZAR	MISHA: reading form
SHIFT	±7 degrees	MISHA: extra interval per pass
MAX INT	1–12 degrees	SAMCHILL: maximum interval
PESO (weight)	0–100 %	SAMCHILL: preference for small intervals
BORDE (edge)	REBOTE, ENVOLVER	Bounce back or wrap around when leaving the register
BASE / RANGO / DIV		Register and step
PROB	0–100 %	Probability that the step plays (the interval accumulates anyway)
OCT	0–100 %	Octave displacements

Catalogue of the 256 presets

The presets are starting points, not exact reconstructions: each one translates the technique or central idea of a work into Aurelia's engines. The preset names stay as they appear in the interface. The *Voices* column gives the mode of each active voice; (CCnn) marks a voice that sends that controller instead of notes.

1. Xenakis and stochastic music

No.	Preset	Inspiration and technique	Voices
1	**Metastaseis**	Masses of glissandi that open and close, like the strings of Metastaseis (1954): four clouds with GLISS at maximum and unscaled pitches.	ESTOC · ESTOC · ESTOC · ESTOC
2	**Pithoprakta**	Clouds of pizzicati whose velocities follow the Gauss law, like the gas molecules of Pithoprakta (1956). Very short notes, high density.	ESTOC · ESTOC · ESTOC · ESTOC
3	**Achorripsis**	The Achorripsis matrix (1957): each voice alternates active and silent sections following Poisson, with different densities per voice.	ESTOC · ESTOC · ESTOC · ESTOC
4	**Analogique A**	The Markov chains of screens of Analogique A (1958) with the original GTPM matrix, in three string registers.	ANALOG · ANALOG · ANALOG
5	**Analogique A acoplado**	Analogique A with the three coupled alpha/beta chains and a protocol of sections perturbed towards screen H.	ANALOG · ANALOG · ANALOG
6	**ST/10**	Computer-calculated stochastic music like ST/10 (1962): short sections with uniform laws and four timbres that come and go.	ESTOC · ESTOC · ESTOC · ESTOC
7	**ST/4 cuarteto**	Stochastic string quartet in the manner of ST/4: cello, viola and two violins with medium-density clouds in their registers.	ESTOC · ESTOC · ESTOC · ESTOC
8	**Herma**	Piano in the manner of Herma (1961): fast clouds that sweep the whole keyboard with the linear interval law.	ESTOC · ESTOC
9	**Eonta**	Piano bursts against long brass notes, as in Eonta (1964).	ESTOC · ESTOC · ESTOC
10	**Nomos Alpha**	The cube rotations of Nomos Alpha (1966) as permutations of an 8-note row on the cello.	SERIAL · ESTOC
11	**GENDY3**	Four second-order dynamic stochastic polygons, microtonal and gliding, like the layers of GENDY3 (1991).	GENDYN · GENDYN · GENDYN · GENDYN
12	**S.709**	Fast, rough first-order random walks, as in S.709 (1994), the last GENDYN work.	GENDYN · GENDYN

No.	Preset	Inspiration and technique	Voices
13	**GENDY3 lento**	GENDYN at the scale of form: segments several beats long, narrow barriers and long glissandi drawing moving drones.	GENDYN · GENDYN · GENDYN
14	**Concret PH**	Crackling embers as in Concret PH (1958): thousands of very short, high, microtonal grains.	ESTOC · ESTOC · ESTOC
15	**Bohor**	A thick mass that hardly changes, like Bohor (1962): overlapping clouds of long notes.	ESTOC · ESTOC
16	**Mycenae Alpha**	Freehand UPIC drawings (Mycenae Alpha, 1978): long GENDYN lines with continuous glissando.	GENDYN · GENDYN · GENDYN
17	**Evryali**	Piano arborescences as in Evryali (1973): a melodic trunk that branches into variants.	BLOOM · BLOOM
18	**Mists**	Piano random walks as in Mists (1980): the melody crawls through the register leaving trails.	ESTOC · ESTOC
19	**Psappha (criba)**	Xenakis sieves as in Psappha (1975): a union of residue classes (5 and 7) giving irregular but periodic attacks.	EUCLID · EUCLID · EUCLID
20	**Syrmos**	Slow string glissandi that cross, as in Syrmos (1959).	ESTOC · ESTOC · ESTOC
21	**Terretektorh**	Sound scattered among the players as in Terretektorh (1966): four clouds whose centres drift through the whole register.	ESTOC · ESTOC · ESTOC · ESTOC
22	**Persephassa**	Hits rotating among six percussionists as in Persephassa (1969): shifted Euclidean rings chasing each other.	EUCLID · EUCLID · EUCLID · EUCLID
23	**Ley de Cauchy**	A single voice with the Cauchy law: almost all small steps, but every now and then a huge leap.	ESTOC
24	**Paseo browniano**	Pure Brownian motion: each note comes from the previous one by a random step, in scale.	ESTOC · ESTOC
25	**Nube de Poisson**	The purest form of stochastic music: Poisson attacks and uniform pitches in four registers.	ESTOC · ESTOC · ESTOC · ESTOC
26	**Cribas 3∪5 / 7∪8**	Two Xenakis sieves: the union of periods 3 and 5 and that of 7 and 8, in opposite registers.	EUCLID · EUCLID
27	**Arcoseno**	The arcsine law favours the extremes: the voice jumps between low and high and avoids the centre.	ESTOC · ESTOC
28	**GENDYN encerrado**	Almost closed barriers: the polygon hits narrow walls again and again and the melody vibrates within a small interval.	GENDYN · GENDYN
29	**Granular microtonal**	Microtonal grains in the middle register: dense cloud, no scale and maximum MICRO deviation.	ESTOC · ESTOC · ESTOC
30	**Densidad que respira**	Density drifts to the maximum: the cloud thickens to saturation and empties to silence.	ESTOC · ESTOC
31	**Achorripsis II**	Achorripsis matrix with more active voices and short sections: a mosaic texture.	ESTOC · ESTOC · ESTOC · ESTOC

No.	Preset	Inspiration and technique	Voices
32	**Cuarteto gaussiano**	Four voices with the Gauss law and moderate walk in the acoustic scale: stochastic but consonant.	ESTOC · ESTOC · ESTOC · ESTOC

2. Aleatoric music and indeterminacy

No.	Preset	Inspiration and technique	Voices
33	**Music of Changes**	Cage (1951): pitches, rhythms and dynamics drawn at random from charts, as with the I Ching. PR2 in ALEA for all three parameters.	SERIAL · SERIAL
34	**Imaginary Landscape 4**	Cage (1951), twelve radios: four independent sources that come and go in random sections.	ESTOC · ESTOC · ESTOC · MARKOV
35	**Williams Mix**	Cage (1952): hundreds of spliced tape fragments. Grains of all lengths, dense and cut into sections.	ESTOC · ESTOC · ESTOC
36	**Silencio estructurado**	A tribute to 4'33'': almost everything is silence. Isolated, soft notes that make you hear the space between them.	ESTOC · ESTOC
37	**Atlas Eclipticalis**	Cage (1961): notes taken from star charts. Isolated, long, high points, like stars.	ESTOC · ESTOC · ESTOC · ESTOC
38	**Ryoanji**	Cage (1983): slow glissandi over an irregular percussion pulse that never changes pitch.	ESTOC · ESTOC · EUCLID
39	**Cheap Imitation**	A single melodic line moving by step with randomly chosen intervals, like the single line of Cheap Imitation.	INTERV
40	**HPSCHD**	Cage and Hiller (1969): superimposed layers of musical dice. Four independent, saturated PR2 voices.	SERIAL · SERIAL · SERIAL · SERIAL
41	**Études Australes**	Cage (1974): both hands read star maps. Points scattered over the whole keyboard, with some notes in common.	ESTOC · ESTOC
42	**Number Pieces**	The time brackets of the Number Pieces: long notes entering and leaving within flexible windows.	ESTOC · ESTOC · ESTOC · ESTOC
43	**Projection 1**	Feldman (1950): graphic notation of register (high, middle, low) without fixed pitches. Very soft.	ESTOC · ESTOC · ESTOC
44	**Rothko Chapel**	Feldman (1971): soft, sustained, almost motionless chords, with a viola melody that appears now and then.	SUBSEQ · INTERV
45	**Piano and String Quartet**	Feldman (1985): figures repeated almost identically for a long time, barely changing.	TURING · TURING
46	**December 1952**	Earle Brown: a graphic score of rectangles. Points with random duration and register, all independent.	ESTOC · ESTOC
47	**Available Forms**	Earle Brown (1961): mobile forms reordered by the conductor. Bloom in random order with branches.	BLOOM · BLOOM

No.	Preset	Inspiration and technique	Voices
48	**For 1, 2 or 3 People**	Christian Wolff (1964): the players react to one another. Short voices harmonically fitted to all the others.	ESTOC · ESTOC · ESTOC
49	**Klavierstück XI**	Stockhausen (1956): fragments played in the order the eye finds them. A 19-step trunk read at random.	BLOOM
50	**Tercera Sonata (Boulez)**	Boulez (1957): formants whose order is permuted. Lexicographic permutations of a 7-note row.	SERIAL
51	**Jeux vénitiens**	Lutosławski (1961), controlled aleatoricism: each instrument repeats its cell at its own speed, unsynchronized.	TURING · TURING · TURING · TURING
52	**Cuarteto (Lutosławski)**	Lutosławski (1964): mobiles of repeated cells of different lengths, superimposed by the ensemble.	TURING · TURING · TURING · TURING
53	**Juego de dados (Mozart)**	The Musikalisches Würfelspiel attributed to Mozart (1792): bars chosen by dice over a fixed harmony.	SUBSEQ · SUBSEQ
54	**Polonesas (Kirnberger)**	Der allezeit fertige Polonoisen- und Menuettencomponist (1757): a dance composed by dice, in AABA form.	SUBSEQ · SUBSEQ
55	**Zyklus**	Stockhausen (1959): a percussionist travels round a cycle of instruments. Rings with XOR logic and rotating hits.	EUCLID · ESTOC
56	**Mobile (Calder)**	A mobile form that swings back and forth, like a Calder mobile: Bloom in pendulum order with changing branches.	BLOOM · BLOOM
57	**Treatise**	Cardew (1967): a graphic score of continuous lines. Two slow GENDYN lines and scattered points.	GENDYN · GENDYN · ESTOC
58	**The Great Learning 7**	Cardew (1968), paragraph 7: each voice picks a note it hears and holds it; a choir of long notes emerges.	ESTOC · ESTOC · ESTOC · ESTOC
59	**Variations II**	Cage (1961): points and lines on transparencies. Fully uniform values in pitch, dynamics and duration.	ESTOC · ESTOC
60	**Palais de Mari**	Feldman (1986): small piano figures that come back again and again, with minimal variations.	TURING · TURING
61	**Dados por parámetro**	Each parameter rolls its own die: pitches, rhythm and dynamics fully random (PR2 in ALEA) in four voices.	SERIAL · SERIAL · SERIAL · SERIAL
62	**Cara o cruz**	The coin as generator: a Turing register with BLOQUEO at 50 % is pure chance, bit by bit.	TURING · TURING
63	**I Ching (hexagramas)**	Six lines, like a hexagram: a 6-bit register that changes some lines on each reading.	TURING · TURING
64	**Indeterminación (ventanas)**	Controlled aleatoricism by sections: the material is fixed, but when each voice plays is random.	SUBSEQ · ESTOC · ESTOC

3. Serialism

No.	Preset	Inspiration and technique	Voices
65	**Suite op. 25**	Schoenberg (1923), the first fully twelve-tone work: the row in its four forms, one after another.	SERIAL · SERIAL
66	**Sinfonía op. 21**	Webern (1928): a canon in which one voice reads the row and the other reads it backwards (palindrome).	SERIAL · SERIAL
67	**Concierto op. 24**	Webern (1934): a row derived from a three-note cell, transformed (P, RI, R, I) and transposed.	SERIAL
68	**Variaciones op. 27**	Webern (1936): mirror symmetry. One voice plays the row and another its inversion, at the same time.	SERIAL · SERIAL
69	**Mode de valeurs**	Messiaen (1949): each pitch has its own duration and dynamic. PR1 at the irregular level for all three parameters.	SERIAL · SERIAL
70	**Structures Ia**	Boulez (1952), integral serialism for two pianos: strict row in pitches, durations and dynamics.	SERIAL · SERIAL
71	**Kreuzspiel**	Stockhausen (1951): crossing registers. One voice goes down while the other goes up.	SERIAL · SERIAL · EUCLID
72	**Gruppen**	Stockhausen (1957): three orchestras at different tempi. Three rows with units of 1/16, 1/16T and 1/8T.	SERIAL · SERIAL · SERIAL
73	**Il canto sospeso**	Nono (1956): the row is shared note by note among the voices (pointillist hocket).	SERIAL · SERIAL · SERIAL
74	**Combinatoriedad (Babbitt)**	Babbitt: combinations of the row. Hexachords combined without repeating pitches.	SERIAL · SERIAL
75	**Rotación (Krenek)**	Krenek: the row rotates one step each pass and generates its own variants.	SERIAL · SERIAL
76	**Interversiones (Messiaen)**	The interversions of Chronochromie (1960): successive permutations of an order of values.	SERIAL · SERIAL
77	**Übung für Klavier (PR1)**	Koenig (1970), Project 1: almost irregular pitches, regular rhythm and intermediate dynamics.	SERIAL
78	**PR1 de regular a irregular**	Three PR1 voices at levels 2, 4 and 6: from almost strict row to almost total chance.	SERIAL · SERIAL · SERIAL
79	**PR2 tendencias**	Koenig, Project 2: tendency masks that open and close in pitch, rhythm and dynamics.	SERIAL · SERIAL
80	**PR2 grupos**	Koenig, Project 2: the GRUPO method repeats each chosen value several times in a row.	SERIAL
81	**Funktionen (Koenig)**	Control functions in the manner of Koenig: PR2 with RATIO and SECUENC., fine rhythm and wide register.	SERIAL · SERIAL
82	**Suite lírica (Berg)**	Berg (1926): expressive twelve-tone writing with tonal echoes. Row in harmonic minor, slow.	SERIAL · SUBSEQ
83	**Quaderno musicale**	Dallapiccola (1952): a lyrical, singable serialism with short figures.	SERIAL

No.	Preset	Inspiration and technique	Voices
84	**Rotaciones (Stravinsky)**	The rotation arrays of late Stravinsky: hexachords that rotate and transpose.	SERIAL · SERIAL
85	**Serie dodecafónica**	A 12-note row played in P form, without transformations: the starting point of everything.	SERIAL
86	**Hexacordos combinatorios**	Combinations of 3 out of 6: the material of the hexachord spread over all its subsets.	SERIAL · SERIAL
87	**Puntillismo**	Webern's style: isolated notes in widely separated registers, with plenty of air between them.	SERIAL
88	**Serialismo integral x4**	Four PR1 voices with mixed levels: the texture of 1950s integral serialism.	SERIAL · SERIAL · SERIAL · SERIAL
89	**Valores añadidos**	Messiaen's rhythm: irregular durations (PR1 rhythm level 7) over a regular pitch row.	SERIAL
90	**Isorritmo (Machaut)**	Color and talea of different lengths, as in isorhythmic motets: note loop of 13 and rhythm loop of 7.	ESPAC · ESPAC
91	**Canon cancrizans**	Crab canon, as in Bach's Musical Offering: the same line forwards and backwards.	SERIAL · SERIAL
92	**Fibonacci (Nono, Bartók)**	Fibonacci proportions, frequent in Nono and Bartók: the Fibonacci word decides the steps of the melody.	FRACTAL · FRACTAL
93	**Mantra (Stockhausen)**	Stockhausen (1970): formula composition for two pianos. The formula and its inversion, expanding.	SERIAL · SERIAL
94	**Tierkreis (Stockhausen)**	Stockhausen (1975): twelve tonal zodiac melodies for music box.	SERIAL · SERIAL
95	**Kontra-Punkte**	Stockhausen (1953): points that gradually group together. PR1 with regular levels and groups.	SERIAL · SERIAL
96	**Polifonía serial libre**	Four PR2 voices with different methods: a non-strict serial polyphony.	SERIAL · SERIAL · SERIAL · SERIAL

4. Minimalism and process

No.	Preset	Inspiration and technique	Voices
97	**In C**	Riley (1964): each player repeats a cell and moves to the next when they like, over an eighth-note pulse on C.	BLOOM · BLOOM · BLOOM · EUCLID
98	**Piano Phase**	Reich (1967): two pianos with the same 12-note pattern. Here one has 12 steps and the other 11, so they drift out of phase.	TURING · TURING
99	**Clapping Music (aprox.)**	Reich (1972): the same rhythm in two voices, one shifted against the other (Euclidean approximation).	EUCLID · EUCLID
100	**Music for 18 Musicians**	Reich (1976): chords pulsing in eighth notes and marimba patterns repeating over them.	SUBSEQ · TURING · TURING · SUBSEQ

No.	Preset	Inspiration and technique	Voices
101	**Drumming**	Reich (1971): bongos built up by substituting beats for rests, in hocket.	EUCLID · EUCLID
102	**Electric Counterpoint**	Reich (1987): looped guitar layers that interlock.	TURING · TURING · TURING · SUBSEQ
103	**Six Pianos**	Reich (1973): six pianos with short interlocking patterns in a modal scale.	TURING · TURING · TURING
104	**A Rainbow in Curved Air**	Riley (1969): fast modal organ arpeggios that overlap and mutate slowly.	TURING · BLOOM · TURING
105	**Poppy Nogood (eco)**	Riley: the time-lag accumulator multiplies a phrase into echoes. Here, the same sequence at three lengths.	BLOOM · BLOOM · BLOOM
106	**Proceso aditivo (Glass)**	Glass: arpeggiated figures that grow and shrink. Arpeggios in three different Euclidean rhythms.	SUBSEQ · SUBSEQ · SUBSEQ
107	**Einstein on the Beach**	Glass (1976): non-stop fast arpeggios and a counting bass.	SUBSEQ · SUBSEQ
108	**Music for Airports**	Eno (1978): tape loops of incommensurable lengths. Each voice repeats one note in a cycle of 19, 23, 29 or 31 steps.	EUCLID · EUCLID · EUCLID · EUCLID
109	**Discreet Music**	Eno (1975): two short looped phrases, slow, overlapping differently on each pass.	TURING · TURING
110	**Thursday Afternoon**	Eno (1985): sparse, slow piano over a soft cloud.	ESTOC · ESTOC
111	**Für Alina**	Pärt (1976), tintinnabuli: a melodic voice moving by step and another that only plays notes of the tonic triad.	INTERV · SUBSEQ
112	**Spiegel im Spiegel**	Pärt (1978): triplet arpeggios on the piano and a melody that rises and falls by step.	SUBSEQ · INTERV
113	**Canon de mensuración**	As in Tabula Rasa (Pärt) or Ockeghem's canons: the same line at speeds x1, x2 and x4.	INTERV · INTERV · INTERV
114	**Rational Melodies**	Tom Johnson (1982): self-similar melodies built by rules. The Thue-Morse word.	FRACTAL
115	**Counting Duets**	Tom Johnson: counting as a musical process. Two Euclidean rings adding up their pulses.	EUCLID · EUCLID
116	**The Well-Tuned Piano**	La Monte Young: clouds of notes played as drones, in a narrow, low register.	ESTOC · ESTOC
117	**Trio for Strings**	La Monte Young (1958): extremely long notes that slowly come and go.	GENDYN · GENDYN · GENDYN
118	**Proceso gradual**	A long sequence that changes very slowly: Reich's idea of music as a gradual process.	TURING · SUBSEQ
119	**Arpegiador minimalista**	A modal arpeggio moving with the harmony, over a bass and a pad.	SUBSEQ · SUBSEQ · SUBSEQ

No.	Preset	Inspiration and technique	Voices
120	**Gamelan (kotekan)**	Balinese kotekan: two interlocking voices, one on the beats and one off the beats, in pelog scale.	EUCLID · EUCLID · EUCLID
121	**Hoquetus**	The medieval hocket technique: a melody shared note by note between two voices.	EUCLID · EUCLID
122	**Pulso**	The steady pulse of Riley and Reich: repeated eighth notes in three registers, with small variations.	EUCLID · EUCLID · SUBSEQ
123	**Bucles inconmensurables**	Loops of lengths 16, 7, 5 and 3 for trigger, note, octave and duration: it never repeats the same way.	ESPAC · ESPAC
124	**Phrygian Gates**	Adams (1977): fast modal pulses changing mode through gates. Patterns in Lydian.	TURING · TURING
125	**Shaker Loops**	Adams (1978): tremolos and short loops that vibrate.	TURING · TURING · SUBSEQ
126	**Ostinato menor**	Obsessive minor ostinatos in the style of minimalist film scores: arpeggio, bass and a pad.	SUBSEQ · SUBSEQ · SUBSEQ
127	**Nine Bells**	Tom Johnson (1979): nine bells played along systematic paths. Permutations of 9 pitches.	SERIAL
128	**Sustitución de silencios**	A pattern starts almost empty and fills up (and empties) by mutation: the building technique of Drumming.	BLOOM · BLOOM

5. Berlin School and sequencing

No.	Preset	Inspiration and technique	Voices
129	**Phaedra**	Tangerine Dream (1974): a repeating 16-step sequencer, slow mellotron and a filter that moves by itself (CC74 on channel 1).	TURING · SUBSEQ · GENDYN (CC74) · ESTOC
130	**Rubycon**	Tangerine Dream (1975): a sequence that mutates slowly over a dark drone.	TURING · GENDYN · GENDYN (CC74)
131	**Ricochet**	Tangerine Dream (1975): fast sequence, drums and an improvised solo.	TURING · EUCLID · MARKOV · GENDYN (CC74)
132	**Stratosfear**	Tangerine Dream (1976): sequence and acoustic guitar in triplets, with a bright pad.	TURING · SUBSEQ · BLOOM
133	**Force Majeure**	Tangerine Dream (1979): octave sequences driving forward, bass and solo.	TURING · SUBSEQ · MARKOV
134	**Timewind**	Klaus Schulze (1975): a very long sequencer loop, breathing pads and a filter drifting for half an hour.	TURING · SUBSEQ · GENDYN (CC74) · GENDYN (CC1)
135	**Mirage**	Klaus Schulze (1977): icy textures, bells and slow glissandi.	GENDYN · BLOOM · SUBSEQ
136	**Moondawn**	Klaus Schulze (1976): sequencer and drums in a long groove.	TURING · EUCLID · SUBSEQ · GENDYN (CC74)
137	**X**	Klaus Schulze (1978): layered sequences crossing each other, faster and denser.	TURING · TURING · SUBSEQ

No.	Preset	Inspiration and technique	Voices
138	**E2-E4**	Manuel Göttsching (1984): two Dorian chords alternating for an hour, a pulse and an improvising guitar.	TURING · SUBSEQ · MARKOV · EUCLID
139	**New Age of Earth**	Ashra (1976): bright, luminous arpeggios over a major pad.	SUBSEQ · SUBSEQ · BLOOM
140	**Silver Apples of the Moon**	Subotnick (1967) with the Buchla sequencer: strange, jumpy, constantly changing chromatic patterns.	TURING · TURING · GENDYN (CC74)
141	**Departure (Hoenig)**	Michael Hoenig (1978): superimposed sequences of different lengths (Kria polymetre).	ESPAC · ESPAC · SUBSEQ
142	**I Feel Love (bajo)**	Moroder (1977): sixteenth-note sequencer bass with octave jumps, and kick on the beat.	ESPAC · EUCLID · SUBSEQ
143	**Trans Europa Express**	Kraftwerk (1977): metallic, repetitive sequences, like a train.	EUCLID · TURING · EUCLID
144	**Oxygène**	Jarre (1976): a wide pad, a floating arpeggio and a singable melody.	SUBSEQ · SUBSEQ · SUBSEQ · GENDYN (CC1)
145	**Structures from Silence**	Steve Roach (1984): chords that transform extremely slowly.	SUBSEQ · GENDYN · GENDYN (CC74)
146	**Berlin oscuro**	The dark Berlin School of Redshift or Node: sequence in harmonic minor, drone and slow filter.	TURING · GENDYN · GENDYN (CC74) · GENDYN (CC71)
147	**Secuencia + filtro**	The Berlin School basics: 16 looped steps and a sample & hold on the filter (CC74) on the same channel.	TURING · TURING (CC74)
148	**Acid**	TB-303 line: sequence with slides and an erratic filter sweep (CC74) on the same channel.	TURING · GENDYN (CC74) · EUCLID
149	**Mellotron y arpegio**	Mellotron pads and an organ arpeggio: the palette of the early Berlin School.	SUBSEQ · SUBSEQ
150	**Bajo + resonancia**	Bass sequence with resonance (CC71) and cutoff (CC74) moving at different rates.	TURING · TURING (CC71) · TURING (CC74)
151	**Berlin polimétrico**	Three sequencers of 16, 15 and 13 steps: the patterns cross and take a long time to line up again.	TURING · TURING · TURING
152	**Modulación lenta (solo CC)**	Four slow modulations for your synths: CC74 and CC71 on channel 1, CC1 and CC11 on channel 2.	GENDYN (CC74) · GENDYN (CC71) · GENDYN (CC1) · GENDYN (CC11)
153	**Kosmische**	German cosmic music: drone, sequencer echo and spacey glissandi.	TURING · GENDYN · ESTOC
154	**Zuckerzeit (Cluster)**	Cluster/Harmonia (1974): simple, cheerful ostinatos repeating like toys.	TURING · TURING · EUCLID
155	**Motorik**	The motorik rhythm of Neu!: straight drums and a bass that never stops.	EUCLID · SUBSEQ · TURING
156	**Bayreuth Return**	Klaus Schulze: a very fast sequence, almost a vibration, with a breathing filter.	TURING · GENDYN (CC74) · SUBSEQ

No.	Preset	Inspiration and technique	Voices
157	**Turing en mutación**	A sequence that changes by itself on each pass and can be locked when it sounds good (raise BLOQUEO to 100).	TURING · TURING
158	**René en Berlín**	Make Noise's Cartesian sequencer applied to the Berlin School: a 4×4 grid read in zigzag.	ESPAC · ESPAC · GENDYN (CC74)
159	**Doble secuenciador**	Two identical sequences, one in sixteenths and one in triplets, shifting against each other.	TURING · TURING
160	**Pads con expresión**	Long chords with expression (CC11) and modulation (CC1) moving like a hand on the keyboard.	SUBSEQ · GENDYN (CC11) · GENDYN (CC1)

6. Generative and algorithmic

No.	Preset	Inspiration and technique	Voices
161	**Illiac Suite I**	Hiller and Isaacson (1957), experiment 1: first-species counterpoint generated by rules, against a cantus firmus.	TURING · REGLAS
162	**Illiac Suite IV**	Hiller and Isaacson, experiment 4: variable-order Markov chains over a corpus.	MARKOV · MARKOV
163	**Music Mouse**	Laurie Spiegel (1986): a cursor moves two pairs of voices with rules of harmony and contrary motion.	REGLAS
164	**Expanding Universe**	Laurie Spiegel (1980, GROOVE system): slow processes unfolding without hurry.	GENDYN · REGLAS
165	**Bloom (Eno y Chilvers)**	The Bloom app (2008): notes that loop and fade, in a pentatonic scale.	BLOOM · BLOOM
166	**Otomata**	Batuhan Bozkurt (2011): cells bouncing in a grid and sounding when they hit the walls.	AUTÓM
167	**Wolfram Tones (regla 30)**	Wolfram's rule 30: deterministic chaos read as melody.	AUTÓM · AUTÓM
168	**Regla 90 (Sierpinski)**	Rule 90 draws the Sierpinski triangle: fractal patterns that repeat and double.	AUTÓM · AUTÓM
169	**Regla 110**	Rule 110, capable of universal computation: structures that travel and collide.	AUTÓM · AUTÓM
170	**Autómatas en cuarteto**	Four rows of elementary automata with consecutive rules: each birth is a note.	AUTÓM
171	**Voyage (Nørgård)**	Nørgård (1968): the infinity series, a melody that contains slow versions of itself.	FRACTAL · FRACTAL
172	**Sinfonía n.º 2 (Nørgård)**	Nørgård (1970): the infinity series superimposed at several speeds at once.	FRACTAL · FRACTAL · FRACTAL · FRACTAL
173	**Sinfonía n.º 3 (Nørgård)**	The infinity series with an interval of a third and fractal rhythm.	FRACTAL · FRACTAL

No.	Preset	Inspiration and technique	Voices
174	**Ruido 1/f (Voss)**	Voss and Clarke (1975): music resembles 1/f noise. A pink melody, halfway between chance and a walk.	FRACTAL · FRACTAL
175	**Blanco, rosa, marrón**	Voss's three noise colours: white (chance), pink (1/f) and brown (random walk).	FRACTAL · FRACTAL · ESTOC
176	**L-system Fibonacci**	The Fibonacci grammar (A→AB, B→A): a quasi-periodic melody that never repeats exactly.	FRACTAL
177	**Thue-Morse**	The Thue-Morse sequence: perfect balance between rises and falls.	FRACTAL · FRACTAL
178	**Conjunto de Cantor**	Cantor dust: sound and silence in fractal proportions.	FRACTAL · FRACTAL
179	**Planta (Lindenmayer)**	A Lindenmayer plant turned into a score: branches that rise, come back and grow again.	FRACTAL · FRACTAL
180	**OMClouds**	OpenMusic adaptive search: an 8-note phrase corrects itself to meet rules of intervals and leaps.	REGLAS · REGLAS
181	**Situation**	The Situation library (OpenMusic): chords with allowed vertical intervals and minimal movement.	REGLAS
182	**Markov de tu teclado**	Play a melody into the MIDI input: the Markov chain learns it and recombines it.	MARKOV · MARKOV
183	**Ecología de Markov**	The voices learn from one another: a system that listens to itself.	TURING · MARKOV · MARKOV
184	**Subsequence: banda**	The Subsequence library: chord graph, bass, melody and arpeggio listening to each other.	SUBSEQ · SUBSEQ · SUBSEQ · SUBSEQ
185	**René cartesiano**	Make Noise's René: a 4×4 grid with independent X and Y clocks.	ESPAC · ESPAC
186	**Kria polimétrico**	monome's Kria: each parameter with its own loop.	ESPAC · ESPAC
187	**Orca**	In the spirit of Orca: lines of clock, increment and random operators.	ESPAC · ESPAC
188	**Turing Machine**	Music Thing Modular's Turing Machine: a 16-bit register that changes or locks.	TURING · TURING
189	**Turing + Pulses**	The Turing Machine with its Pulses expander: each bit is a gate output.	TURING · TURING
190	**Grids (batería GM)**	The Grids pattern map (Mutable Instruments) playing kick, snare and hi-hat on channel 10.	EUCLID
191	**Samchillian**	Leon Gruenbaum's Samchillian keyboard: each note comes from the previous one by an interval, in automatic mode.	INTERV · INTERV
192	**Misha interválico**	Eventide's Misha: a sequence of intervals cycling through P, R, I and RI.	INTERV · INTERV

7. Contemporary and spectral

No.	Preset	Inspiration and technique	Voices
193	**Continuum**	Ligeti (1968): very fast harpsichord patterns of different lengths that turn into a continuum.	TURING · TURING · TURING
194	**Poème symphonique**	Ligeti (1962), 100 metronomes: pulses at different speeds overlapping.	EUCLID · EUCLID · EUCLID · EUCLID
195	**Atmosphères**	Ligeti (1961): motionless chromatic clusters that change colour very slowly.	ESTOC · ESTOC
196	**Lontano**	Ligeti (1967), micropolyphony: lines moving by semitone in canon at different speeds.	INTERV · INTERV · INTERV · INTERV
197	**Désordre**	Ligeti (1985), étude no. 1: the right hand on white keys and the left chromatic, with shifting accents.	EUCLID · EUCLID
198	**Musica ricercata**	Ligeti (1953): each piece with a growing number of pitches. Here, a row of only 3 notes.	SERIAL
199	**Canon de tempos**	Nancarrow, Studies for Player Piano: the same line at tempi 3:4:6.	INTERV · INTERV · INTERV
200	**Partiels**	Grisey (1975): the harmonic spectrum of a low E orchestrated. Acoustic scale (harmonics 8-14) in open registers.	ESTOC · ESTOC · EUCLID
201	**Una nota sola (Scelsi)**	Scelsi (1959): a single pitch vibrating with microtones, tiny glissandi and changes of intensity.	GENDYN · GENDYN
202	**Threnody**	Penderecki (1960): string clusters and block glissandi.	ESTOC · ESTOC · ESTOC
203	**Spectral CANON (Tenney)**	Tenney (1974), for Nancarrow: each voice is a harmonic and repeats its note at its own rate.	EUCLID · EUCLID · EUCLID · EUCLID
204	**Ergodos**	Tenney: ergodic music, statistically the same at any moment. A stationary uniform distribution.	ESTOC · ESTOC
205	**Batimientos (Lucier)**	Lucier: almost identical sustained notes whose slight detuning produces beating.	ESTOC · ESTOC
206	**Deep Listening**	Pauline Oliveros: slow, deep drones for listening to the space.	GENDYN · GENDYN · GENDYN
207	**Drone (Radigue)**	Éliane Radigue: a single drone changing imperceptibly.	GENDYN
208	**Catalogue d'oiseaux**	Messiaen (1958): birdsong, fast, high and with abrupt leaps.	ESTOC · MARKOV · ESTOC
209	**Modos de Messiaen**	Messiaen: the third mode of limited transposition, with chords and a melody in it.	SUBSEQ · SUBSEQ
210	**Chronochromie**	Messiaen (1960): durations as colour; layers of irregular rhythm overlapping.	SERIAL · SERIAL · ESTOC
211	**Stimmung**	Stockhausen (1968): six voices singing the harmonics of a low B flat.	SUBSEQ · ESTOC · ESTOC

No.	Preset	Inspiration and technique	Voices
212	**Kontakte**	Stockhausen (1960): impulses that speed up into pitch and turn back into rhythm.	GENDYN · ESTOC
213	**Gesang der Jünglinge**	Stockhausen (1956): groups of impulses and clouds with changing densities and registers.	ESTOC · ESTOC · SERIAL
214	**Nueva complejidad**	Ferneyhough and the new complexity: extreme density, gridless rhythms and changing dynamics.	SERIAL · ESTOC · ESTOC
215	**Concreta instrumental**	Lachenmann: noises and dry attacks with extreme dynamics.	ESTOC · ESTOC
216	**Sciarrino (susurros)**	Sciarrino: high, fragile sounds on the edge of silence.	ESTOC · ESTOC
217	**Saariaho (texturas)**	Saariaho: timbral textures changing slowly from noisy to harmonic.	GENDYN · ESTOC
218	**Radulescu (drones)**	Radulescu: interweaving drones of harmonics.	ESTOC · ESTOC · GENDYN
219	**Microtonalidad**	Free pitches between the semitones: MICRO clouds at maximum over a drone.	ESTOC · ESTOC · GENDYN
220	**Ramifications**	Ligeti (1969): two string groups, one tuned a quarter tone apart, in entangled figures.	GENDYN · GENDYN
221	**Fragmentos (Kurtág)**	Kurtág: minimal pieces, brief gestures surrounded by silence.	BLOOM
222	**Ionisation**	Varèse (1931): pure percussion, sirens and hits in irregular rhythms.	EUCLID · ESTOC · EUCLID
223	**Boogie (Nancarrow 3a)**	Nancarrow, study 3a: boogie-woogie for player piano, with a walking bass and fast layers on top.	SUBSEQ · TURING · SUBSEQ
224	**Rhythmicon (Cowell)**	Cowell and Theremin (1931): each harmonic repeats at a rate proportional to its number.	EUCLID · EUCLID · EUCLID · EUCLID

8. Rhythm, Euclidean and modulation

The first 18 are Euclidean rhythms from Toussaint's list. They all use the same template: a bass with the rhythm, a melody advanced by a second ring, and a high sixteenth-note pulse.

No.	Preset	Inspiration and technique	Voices
225	**Tresillo E(3,8)**	E(3,8), the Cuban tresillo: the most widespread Euclidean rhythm in the world (Toussaint, 2005).	EUCLID · EUCLID · EUCLID
226	**Cinquillo E(5,8)**	E(5,8), the Cuban cinquillo, also found in West African traditional music.	EUCLID · EUCLID · EUCLID
227	**Bossa nova E(5,16)**	E(5,16), the bossa nova pattern from Brazil.	EUCLID · EUCLID · EUCLID
228	**Bembé E(7,12)**	E(7,12), the bembé bell of West Africa and Cuba.	EUCLID · EUCLID · EUCLID
229	**Aksak E(4,9)**	E(4,9), an aksak (limping) rhythm from Turkey.	EUCLID · EUCLID · EUCLID
230	**Cumbia E(3,4)**	E(3,4), the basis of cumbia and calypso.	EUCLID · EUCLID · EUCLID
231	**Samba E(7,16)**	E(7,16), a Brazilian samba pattern.	EUCLID · EUCLID · EUCLID

No.	Preset	Inspiration and technique	Voices
232	**Ruchenitza E(3,7)**	E(3,7), the Bulgarian ruchenitza.	EUCLID · EUCLID · EUCLID
233	**Aka E(13,24)**	E(13,24), a pattern of the Aka pygmies of Central Africa.	EUCLID · EUCLID · EUCLID
234	**Sudáfrica E(5,12)**	E(5,12), a South African rhythm.	EUCLID · EUCLID · EUCLID
235	**Khafif-e-ramal E(2,5)**	E(2,5), the Persian khafif-e-ramal rhythm.	EUCLID · EUCLID · EUCLID
236	**Outside Now E(4,11)**	E(4,11), the metre of Frank Zappa's Outside Now.	EUCLID · EUCLID · EUCLID
237	**Promenade E(5,11)**	E(5,11), the irregular metre of the Promenade from Pictures at an Exhibition (Mussorgsky).	EUCLID · EUCLID · EUCLID
238	**York-Samai E(5,6)**	E(5,6), the York-Samai of the Arabic tradition.	EUCLID · EUCLID · EUCLID
239	**Tuareg E(7,8)**	E(7,8), a Tuareg rhythm from the Sahara.	EUCLID · EUCLID · EUCLID
240	**África central E(9,16)**	E(9,16), a Central African pattern.	EUCLID · EUCLID · EUCLID
241	**Nawakhat E(5,7)**	E(5,7), the Arabic nawakhat rhythm.	EUCLID · EUCLID · EUCLID
242	**Aka E(11,24)**	E(11,24), another Aka pattern.	EUCLID · EUCLID · EUCLID
243	**Polirritmo 3:4**	Two combined Euclidean rings: 3 against 4, the most basic polyrhythm.	EUCLID · EUCLID
244	**Polirritmo 5:4 (Ligeti)**	5 against 4, the kind of polyrhythm found in the études of Ligeti and Nancarrow.	EUCLID · EUCLID
245	**Criba XOR**	Two rings combined with XOR: it sounds where one or the other hits, but not both.	EUCLID · EUCLID
246	**Grids techno**	Grids in a straight corner of the map: kick on the beat, snare on 2 and 4 and hi-hat in eighths (channel 10).	EUCLID · TURING
247	**Grids breakbeat**	Grids towards the syncopated corner with chaos: a break that never stops changing.	EUCLID · SUBSEQ
248	**T-1 ratchets**	Euclidean with repeats (ratchets) and harmony, as on the Torso T-1.	EUCLID · EUCLID
249	**Desplazamiento euclídeo**	The same rhythm in three voices with different rotations: the displacement creates the counterpoint.	EUCLID · EUCLID · EUCLID
250	**Ciclo de gong (gamelan)**	Colotomic structure of the gamelan: gong, kenong, kempul and bonang divide the same cycle into 1, 2, 4 and 8.	EUCLID · EUCLID · EUCLID · EUCLID
251	**Adi tala**	An 8-beat cycle like the adi tala of South India, with a 3-in-8 rhythm on top and a drone.	EUCLID · EUCLID · GENDYN
252	**S&H Turing (CC74)**	Modulation: the Turing Machine as a sample & hold of the filter cutoff (CC74) on channels 1 and 2.	TURING (CC74) · TURING (CC74)
253	**Deriva GENDYN (CC74+71)**	Modulation: two GENDYN polygons move the cutoff (CC74) and resonance (CC71) of channel 1.	GENDYN (CC74) · GENDYN (CC71)

No.	Preset	Inspiration and technique	Voices
254	**Paneo browniano (CC10)**	Modulation: the pan (CC10) of the four channels moves like a Brownian particle.	ESTOC (CC10) · ESTOC (CC10) · ESTOC (CC10) · ESTOC (CC10)
255	**Expresión fractal (CC11)**	Modulation: 1/f noise on the expression (CC11) of channels 1 to 4, like a human player.	FRACTAL (CC11) · FRACTAL (CC11) · FRACTAL (CC11) · FRACTAL (CC11)
256	**Kria de modulación (CC1)**	Modulation: polymetric Kria loops on the mod wheel (CC1) and cutoff (CC74) of channel 1, with its sequence.	TURING · ESPAC (CC1) · ESPAC (CC74)

Recipes, tips and bibliography

Recipes

1. **A harmonic ensemble from scratch.** Put one voice in TURING with BLOQUEO 100 as the guide. Add three voices in different modes with ESC, RIT and ARM on and DIST ARM at 4.4. Everything they generate will respect scale, pulse and consonance with the guide.
2. **From strict to free.** Start with SERIAL PR1 at level 1 and raise RI ALT, RI RIT and RI DIN little by little while recording: the piece moves from the row to chance, as in Koenig's experiments.
3. **Berlin School with modulation.** Voice 1 in TURING (16 steps, 1/16, BLOQUEO 100). Voice 2 in GENDYN with CC 74 on channel 1, CENTRO C5, ANCHO 48 and segments of 1 to 4 beats: the bass filter moves by itself. Lower BLOQUEO to 95 so the sequence mutates very slowly.
4. **Teaching a melody.** A voice in MARKOV with FUENTE = ENTRADA and order 2. Play a phrase several times and let the chain recombine it. CONGELA freezes what it has learned.
5. **Playing intervals.** INTERVÁLICO, type SAMCHILL, FUENTE = ENTRADA and the global TRANSP set to OFF. Each key moves the melody relative to where it was: C4 repeats, D4 goes up a tone.
6. **Mensuration canon.** Three voices with the same mode, the same SEMILLA and DIV 1/8, 1/4 and 1/2: the same line at three speeds (preset 113).
7. **Phasing by length.** Two TURING voices with the same seed and BLOQUEO 100, with lengths 12 and 11: they drift apart and meet again every 132 steps (preset 98).
8. **Generative drums.** EUCLÍDEO in GRIDS with NOTAS = 1, channel 10, ESC and ARM off. Move X and Y to go from straight to syncopated and raise CAOS so it changes by itself.

Tips

- **Seeds.** If you like a result, note down the seeds or store it in a memory: with the same settings, the music is identical.
- **RESET AL PLAY** on makes every play start the same way, ideal for editing. Turn it off so the music keeps evolving between takes.
- **GATE above 100 %** overlaps the notes: needed for continuous glissandi in GENDYN and ESTOCÁSTICO.
- **Pitch bend is per channel.** If two voices share a channel and one uses glissandi or microtones, the bend affects both. Use separate channels for microtonal voices.
- **Many notes per second.** Very high densities in four voices can overload a software synthesizer. Lower DENS or NUBE if you hear dropouts.
- **CC voices.** The value is the note's pitch, so the register decides the controller's range. To sweep the whole range use a large ANCHO or RANGO and ESC off (otherwise the value jumps along the degrees of the scale).

Bibliography

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